

$$V_{\text{rwr supply}} = 30V$$

$$I_{\text{limit}} = 5A$$

$$I_{\text{nom}} = 1A$$

$$P_{\text{nom}} = 30W$$

$$P_{\text{in}} = 30W$$

$$\vec{V}_L = D_A V_{in} - D_B V_{out}$$

$$\text{steady state} \Rightarrow D_A V_{in} = D_B V_{out}$$

$$\frac{V_A}{V_B} = \frac{D_A}{D_B}$$

$$I_{in} \approx D_A I_L$$

$$I_{out} \approx D_B I_L$$

Buck mode.

$$V_{in} > V_{out}$$

$$D_B = 1$$

$$D = \frac{V_{out}}{V_{in}}$$

$$V_{out} = D V_{in}$$

$$V_{L, on} = V_{in} - V_{out} \Rightarrow \text{inductor diagram with } V_{in} \text{ and } V_{out}$$

$$V_{L, off} = -V_{out} \Rightarrow \text{inductor diagram with } -V_{out}$$

$$(V_{in} - V_{out})D + (-V_{out})(1-D) = 0$$

$$\Delta I_L = \frac{(V_{in} - V_{out})D}{L f_s}$$

$$L = \frac{(V_{in} - V_{out})D}{f_s \Delta I_L}$$

$$\Delta I_L \approx \frac{V_{out}(1-D)}{L f_s}$$

$$I_{L, avg} \approx I_{out}$$

Boost mode

$$V_{in} < V_{out}$$

$$D = 1 - \frac{V_{in}}{V_{out}}$$

$$V_{out} = \frac{V_{in}}{1-D}$$

$$V_{L,on} = V_{in}$$

$$V_{L,off} = V_{in} - V_{out}$$

$$V_{in}D + (V_{in} - V_{out})(1-D) = 0$$

$$\Delta I_L = \frac{V_{out}(1-D)}{L f_s}$$

$$L = \frac{V_{in}D}{f_s \Delta I_L}$$

$$I_{L,avg} \cong I_{in}$$

$$\eta = \frac{P_{out}}{P_{in}} = \frac{V_{out} I_{out}}{V_{in} I_{in}}$$

$$T_s = \frac{1}{f_s}$$

$$T_s = \frac{1}{100,000 \text{ Hz}} = 10 \mu\text{s}$$

$$T_s = \frac{1}{50,000 \text{ Hz}} = 20 \mu\text{s}$$

$$V_L = L \frac{dI_L}{dt}$$

$$\Delta I_L = \frac{V_L \Delta t}{L}$$

$$L = \frac{V_L \Delta t}{\Delta I_L}$$

$$\Delta I_L = r \underline{I_{L,avg}}$$

$$r = [0.2, 0.4]$$

$$I_{L,avg} \approx \frac{V_{out} + I_{out}}{\eta V_{in}}$$

$$I_{L,pk} = \underline{I_{L,avg}} + \frac{\Delta I_L}{2}$$

valley current:

$$I_{L,min} = \underline{I_{L,avg}} - \frac{\Delta I_L}{2}$$

$$I_{sat} > I_{L,pk}$$

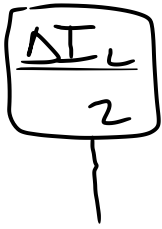
$$I_{sat} \geq [1.25, 1.5] I_{L,pk}$$

CCM

$$I_{L, \min} > 0$$

$$I_{L, \text{avg}} > \frac{\Delta I_L}{2}$$

CCM | DCM



$$E_L = \frac{1}{2} L I_{L, \text{pk}}^2$$

Inductor energy

$$P_{L, \text{cu}} = I_{L, \text{rms}}^2 R_{\text{DCR}}$$

$P = \text{loss}$

$$I_{L, \text{rms}} = \sqrt{I_{L, \text{avg}}^2 + \frac{\Delta I_L^2}{12}}$$

$$P_{\text{core}} = k f_s^\alpha (\text{DB})^{\frac{1}{2}} V_{\text{core}}$$

Output Cap - Buck mode

$$C_{\text{out}} \geq \frac{\Delta I_L}{8 f_s \Delta V_{\text{out}}}$$

$$\Delta V_{\text{out}} \approx \frac{\Delta I_L}{8 f_s C_{\text{out}}} + \Delta I_L \cdot ESR$$

Out cap - boost mode

$$\Delta V_{out} \approx \frac{I_{out} D}{C_{out} f_s}$$

$$C_{out} \geq \frac{I_{out} D}{f_s \Delta V_{out}}$$

Buck out-cap

$$I_{out, rms} \approx \frac{\Delta I_L}{2\sqrt{3}}$$

Boost out-cap

$$I_{out, rms} \approx I_{out} \sqrt{\frac{D}{1-D}}$$

Buck in-cap

$$I_{in, rms} \approx I_{out} \sqrt{D(1-D)}$$

$$\text{at } D = 0.5 \quad I_{in, rms} = I_{out} / 2$$

Buck in ΔV_{in} (ripple)

$$\Delta V_{in} \approx \frac{I_{out} D(1-D)}{C_{in} f_s}$$

$$C_{in} \geq \frac{I_{out} D(1-D)}{f_s \Delta V_{in}}$$

$$P_{cap, ESR} = I_{c, rms}^2 ESR$$

$$V_{DS, rated} > V_{DS, max} \quad V_{DS, rated} \approx 1.5 V_{max}$$

$$I_{FET, peak} > I_{L, peak}$$

$$P_{conduction} = I_{rms}^2 R_{DS(on)}$$

$$P_{cond.} \approx I_L^2 R_{DS(on)} D_x$$

$$\text{Bus input} \rightarrow P_{HS} \approx I_L^2 R_{DS(on)} D$$

$$\rightarrow P_{LS} \approx I_L^2 R_{DS(on)} (1-D)$$

$$R_{DS(on), hot} = R_{DS(on) 25^\circ C} \times K_T$$

$$P_{sw} \approx \frac{1}{2} V_{DS} I_D (t_r + t_f) f_s$$

$$P_{Coss, FET} \approx \frac{1}{2} C_{oss} V_{DS}^2 f_s$$

$$P_{Coss, FET} = E_{oss} f_s$$

$$I_{gate, avg} = Q_g f_s$$

$$P_{gate} = Q_g V_{GS} f_s$$

$$P_{gate, total} = \sum Q_{gi} V_{Gsi} f_s$$

$$= 4 Q_g V_{GS} f_s \text{ in synchronous}$$

Gate drive supply current

$$I_{drive} \approx Q_g f_s + I_q \rightarrow \text{gate driver quiescent current}$$

cont.

$$P = Q_g V_{GS} f$$

$$f_{max, s} \approx \frac{P_{available}}{Q_g V_{GS}}$$

Gate peak current during miller plateau

$$I_g \approx \frac{V_{driver} - V_{plateau}}{R_{drive} + R_{g, ext} + R_{g, int}}$$

Muller transition time

$$t_{miller} \approx \frac{Q_{gd}}{I_g}$$

$$R_g \uparrow \Rightarrow I_g \downarrow \Rightarrow t_{sw} \uparrow \Rightarrow P_{sw} \uparrow$$

$$R_{g, ext} \\ I_g \approx \frac{Q_{gd}}{t_{sw}}$$

$$R_{total} \approx \frac{V_{driver} - V_{plateau}}{I_g}$$

$$R_{g, ext} = R_{total} - R_{driver} - R_{g, int}$$

Gate pull-down R current

$$I_{pull} = \frac{V_{GS}}{R_{GS}}$$

$$P_{R_{GS}} = \frac{V_{GS}^2}{R_{GS}}$$

Mosfet loss

$$P_{FET} = P_{cond.} + P_{sw} + P_{E_{oss}} + P_{diode}$$

$$P_{diode} \approx V_F I_L (2t_{dead} f_s)$$

$$t_{dead} > t_{MOSFET, off} + t_{delay, mismatch}$$

$$t_{dead} \ll T_s$$

Duty
lost
to dead time

$$D_{dead} \approx 2t_{dead} f_s$$

Min pulse width

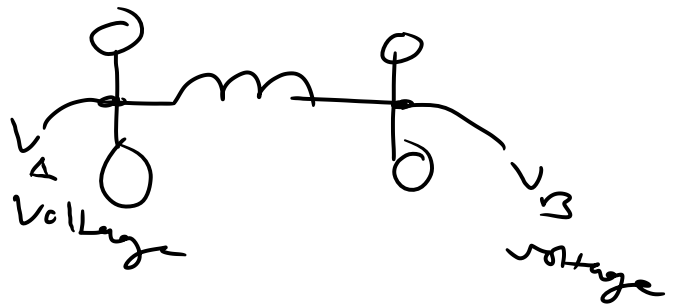
$$t_{on} = D T_s = \frac{D}{f_s}$$

$$t_{off} = (1-D) T_s$$

Max practical duty cycle
 $D_{max} \leq 1 - t_{off, min} f_s$

$$D_{min} \geq f_s t_{on, min}$$

$$L \frac{di_L}{dt} = V_A - V_B$$



$$\frac{di_L}{dt} = D_A V_{in} - D_B V_{out}$$

$$L$$

$$\frac{di_L}{dt} = 0 \text{ at steady state}$$

Buck state equations

$$\frac{di_L}{dt} = \frac{DV_{in} - V_{out}}{L}$$

$$\frac{dV_{out}}{dt} = \frac{i_L - V_{out}/R}{C_{out}} \quad \rightarrow \text{resistive load}$$

Boost avg state space eq

$$\frac{di_L}{dt} = \frac{V_{in} - (1-D)V_{out}}{L}$$

$$\frac{dV_{out}}{dt} = \frac{(1-D)i_L - V_{out}/R}{C_{out}}$$

Load resistance

$$R_{\text{Load}} = V_{\text{out}} / I_{\text{out}}$$

$$R_{\text{Load}} = V_{\text{out}}^2 / P_{\text{out}}$$

$$P_R = V_{\text{out}}^2 / R_{\text{Load}}$$

Junction temp

$$T_J = T_A + P_{Fet} \theta_{JA}$$

θ_{JA} ambient

$$T_J = T_A + P(\theta_{JC} + \theta_{CS} + \theta_{SA})$$

$$T_J < T_{J,max}$$

$$P_{loss, total} = P_{Fet} + P_L + P_C + P_{driver} + P_{control}$$

$$\eta = \frac{P_{in}}{P_{out} + P_{loss, total}}$$

$$|V_{in} - V_{out}| \leq \Delta V$$

$$\Delta V \in [0.5, 1]$$

$$D_A = D_B \quad \text{transition zone}$$

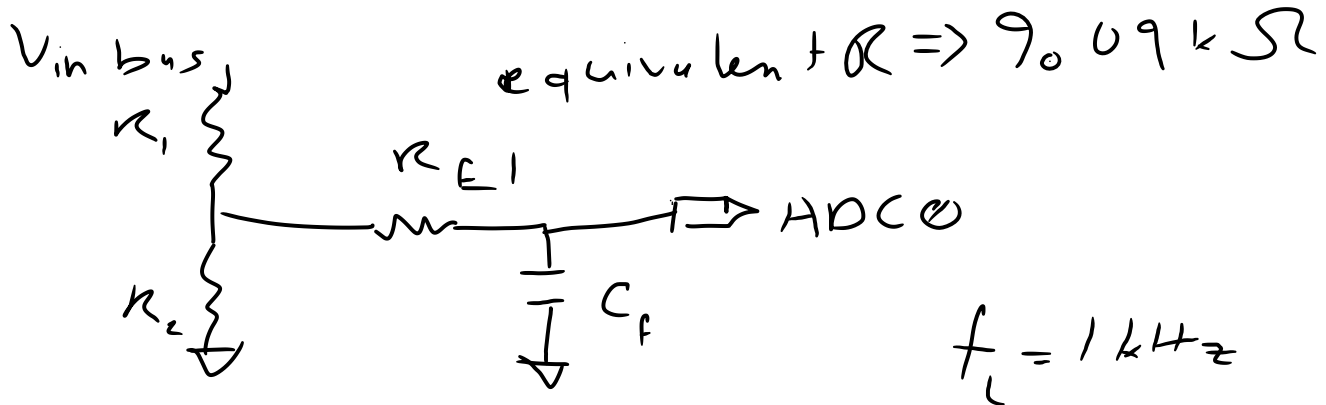
$$D_A = D_B = D_x \text{ ex. } 50\% \text{ or } 70\%$$

$$V_L = V_A - V_B = 0$$

main

- init ADC
 - PWM
 - init the memory of Duty Cycle
 - init compare blocks
 - generate triangle wave A
 - ↳ output wave shrinks and grows w/ D
 - generate triangle wave B
 - ↳ shrinks and grows w/ (1-D)
- ← V_{bus} in AD
- ← V_{bus} out ADC
- array, ramp from $D = 0.3 \rightarrow D = 0.8$
 - gate behavior state machine
 - ↳ hysteresis mode (transition)
 - ↳ boost
 - ↳ buck
 - overall interrupt service CPU timing triangle wave
 - ADC0, ADC1 being watched for overspike and short for enable
 - ↳ highest frequency.
 - ↳ watchdog?

$$\Delta T_L = \frac{(30 - 20) 0.4}{(5572-6)(50,000)} = \frac{40}{...}$$



$$f_L = 1 kHz$$

$$f_c = \frac{1}{2\pi RC}$$

$$C = 18 nF$$

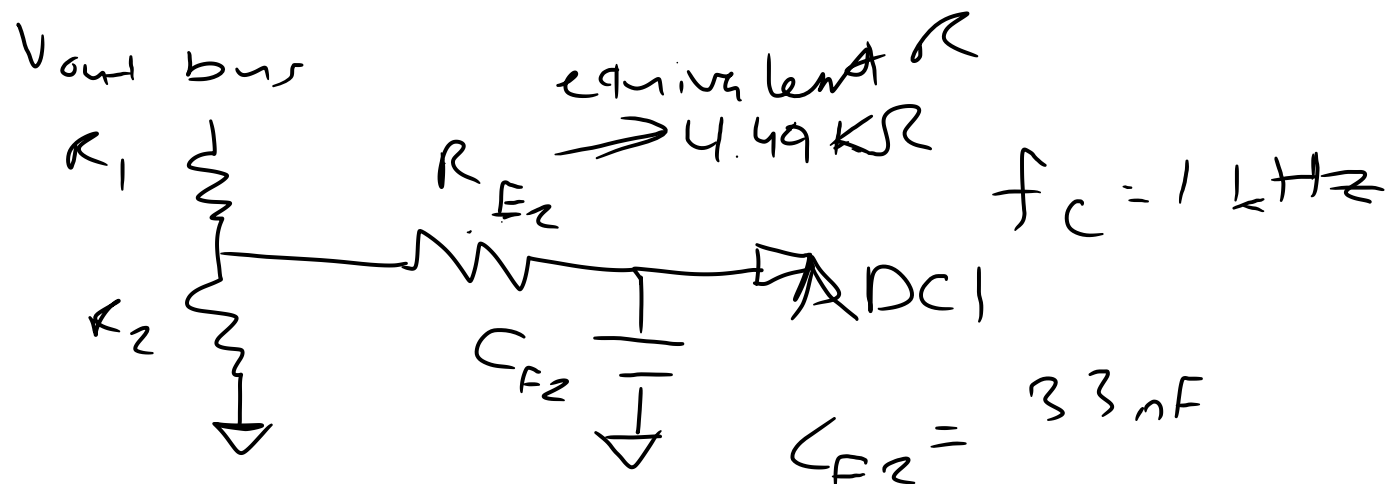
$$3.3V \Rightarrow 36.3V$$

$$3.3 = 36.3 \left(\frac{R_2}{R_1 + R_2} \right)$$

$$3.3R_1 + 3.3R_2 = 36.3R_2$$

$$R_2 = 10 k\Omega$$

$$R_1 = 100 k\Omega$$



$$f_c = 1 kHz$$

$$C_{F2} = 33 nF$$

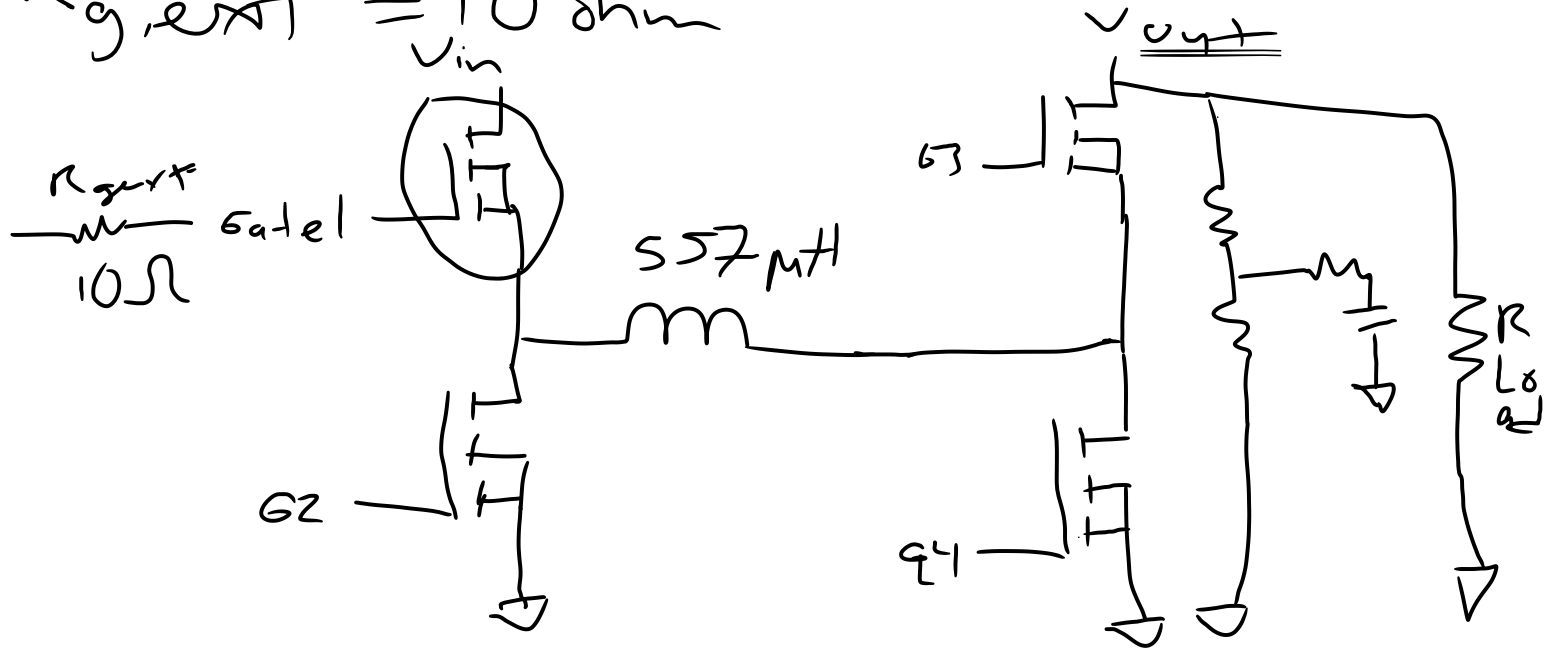
$$3.3V \Rightarrow 73.5V$$

$$R_1 = 100 k\Omega$$

$$R_2 = 4.7 k\Omega$$

$$f_s = 50 kHz$$

$$R_{g, ext} = 10 \Omega$$



$$R_{load} = 6.8 k\Omega$$

$$P_{load max} = 1 W$$

$$V = IR \quad \frac{V}{R} = I$$

$$P = \frac{V^2}{R}$$

$$P = \frac{60 V^2}{6.8 k\Omega} = 0.53 W$$

$$Q_{g max} = 234 nC$$

$$Q_{g nom} = 156 nC$$

$$\text{MER1 ReLoad } R = 2 k\Omega$$

$$R_{load} = 120 \Omega \quad I_L = 1.65 A$$

$$= 6.8 k\Omega \quad P_R = 0.63 W$$

$$R_{\text{load}} = 120 \Omega \quad I_L = 1.65 \text{ A}$$

$$= 6.8 \text{ k}\Omega \quad P_R = 0.63 \text{ W}$$

Buck

$$D_A = \frac{V_{\text{out}}}{V_{\text{in}}}$$

$$D_B = 1$$

$$I_L = I_{\text{out}}$$

$$\Delta I_L = \frac{(V_{\text{in}} - V_{\text{out}}) D_A}{L f_s} \quad \text{Input side}$$

$$\Delta V_C = \frac{\Delta I_L}{8 f_s C}$$

Output side

Boost

$$D_A = 1$$

$$D_B = V_{\text{in}} / V_{\text{out}}$$

$$I_L = I_{\text{out}} / D_B$$

$$\Delta I_L = \frac{V_{\text{in}} D}{L f_s}$$

$$I_{\text{rms}} = \frac{\Delta I_L}{2\sqrt{3}}$$